

GET STARTED

1. Flash the firmware to a Cardputer:

```
esptool --chip esp32s3 write_flash 0x0 studio-merged.bin
```

2. Insert a FAT32 microSD card

3. Put 16-bit WAVs in `/samples` for the sampler

Synth & sequencer work without an SD card. Recording / sampling / saving need one.

MAIN MENU

- 1 SYNTH** sequencer + drums + phrase
- 2 SAMPLER** play a WAV across the keys
- 3 RECORDER** record the mic to SD
- 4 SONGS** browse / preview / load
- 5 LIBRARY** browse / play recordings
- ``` back to menu, anywhere

SAMPLER & RECORDER

SAMPLER `1-8` pick `,` `.` prev/next `tab` page
note keys play the WAV pitched

RECORDER `space` rec/stop `ok` play back

SONGS/LIB `;` `.` move `/` rename `del` delete

NOTE KEYS

`z s x d c v g b h n j m`
`q w e r t y u i o p`

Two rows = two octaves, like a piano (white + black keys).

SYNTH — PLAY & EDIT

Play `enter` play/stop `space` stop

Edit `tab` track `,` `.` cursor `del` clear

Tempo `[` `]` bpm `fn` + `[` `]` loop length

Song `-` `=` slot `/` save `\` load

SYNTH — TRACKS A / B

Voice `1-4` wave `0` use a recording

Level `;` `'` volume `fn` + `;` `'` tone

FX `5` DLY `6` OCT `7` SUB `8` 5TH `9` CHO

Chord `fn` +note C Dm Em F G Am

Chance `fn` + `,` `.` step probability

DRUM TRACK

Hits note keys = 8 drums

Kit `0` ACOU / 808 / 909 `;` `'` volume

PHRASE TRACK — plays a slice of a recording

Source `0` pick recording `z` preview

Per step `1-4` OCT SUB CHO DLY `9` rev

`5` `6` offset `7` `8` length

`fn` + `1` `2` pitch `fn` + `9` stutter

Trim `fn` + `5` `6` start `fn` + `7` `8` segment

LIVE FX

`f` crush `k` tape-stop `a` arp `l` reverse

`fn` + `k` swing `fn` + `l` reverse sequence

Tip: inside SYNTH press `ctrl` for the full key map on the device.